

Year 8 Mathematics  
Mini Test  
Percentages and decimals



Baldivis  
Secondary College

Name: SOLUTIONS

Time: 35 mins

Resources:

/ 43 mark

1 x A4 page of hand written notes (one sided)  
Scientific Calculator

**Part A – Multiple Choice** – Circle only 1 answer

**9 marks**

↓ Use this space for your working ↓

**Question 1**

What is 50% of \$220?

- ☒ a) \$110
- b) \$270
- c) \$170
- d) \$50

$$\frac{50}{100} \times 220 \quad \checkmark$$

**Question 2**

Which of the following is 10% of \$220?

- a) \$11
- b) \$10
- c) \$230
- ☒ d) \$22

$$\frac{10}{100} \times 220 \quad \checkmark$$

**Question 3**

If I increase \$45 by 10% I get which of the following

- a) \$49
- ☒ b) \$49.50
- c) \$55
- d) \$50.50

$$\frac{110}{100} \times 45 \quad \checkmark$$

**Question 4**

If I decrease \$36 by 50% I get which of the following

- a) -\$14
- b) \$86
- c) \$61
- ☒ d) \$18

$$\frac{50}{100} \times 36 \quad \checkmark$$

### Question 5

If my cost price is \$22 and my selling price is \$31, what is my profit?

e) \$11

f) -\$9

g) -\$11

☒ h) \$9

$$\begin{array}{r} \$31 \\ - \$22 \\ \hline \end{array} \quad \checkmark$$

### Question 6

450 reduced by 20% is;

a) 540

b) 405

☒ c) 360

d) 430

$$\frac{80}{100} \times 450 \quad \checkmark$$

### Question 7

Which of the follow is a recurring decimal.

a)  $\frac{1}{4}$

b)  $\frac{10}{25}$

c)  $\frac{1}{8}$

☒ d)  $\frac{1}{3}$

$$0.\dot{3}\dot{3} \quad \checkmark$$

### Question 8

If I bought a painting for \$150 and sold it for \$250 dollars, what is my percentage profit?

a) 100%

☒ b) 66%

c) -66%

d) -100%

$$\begin{array}{l} \checkmark \quad CP = \$150 \\ \quad SP = \$250 \\ \quad P = \$100 \\ \quad \frac{100}{150} \times 100 \end{array}$$

### Question 9

12% Of 250 is;

a) 24

☒ b) 30

c) 50

d) 15

$$\frac{12}{100} \times 250 \quad \checkmark$$

Question 10

(6 marks)

Draw a line to match the term to the definition

| Term          | Definition                                            |
|---------------|-------------------------------------------------------|
| Terminating   | The price you sell an object for. ✓                   |
| Recurring     | A decimal number that repeats continuously forever. ✓ |
| Percent       | How many pieces out of 100 ✓                          |
| Selling price | Amount of money after selling ✓                       |
| Cost Price    | The amount an object costs ✓                          |
| Profit        | A decimal that stops (doesn't continue forever) ✓     |

Question 11

(6 marks)

Write T for terminating and R for recurring next to each of the following decimals.

- a) 0.166666666..... R ✓  
 b)  $0.\overline{538461}$  R ✓  
 c) 1.142857 R ✓

- d) 0.27 T ✓  
 e)  $1.\dot{3}$  R ✓  
 f) 0.00125 T ✓

Question 12

(6 marks)

The following divisions make recurring decimals. Write them in exact form using an appropriate notation.

a)  $2 \div 3 = 0.\dot{6}$  ✓

b)  $5 \div 12 = 0.4\dot{1}\dot{6}$  ✓

c)  $6 \div 7 = 0.\overline{857142} / 0.\dot{8}5714\dot{2}$

1 mark for each calculation  
 1 mark for correct notation.

**Question 13**

(6 marks)

Michelle bought 10 shirts for \$15 each.

- a) If she sells 8 of them for \$20 each, what is the percentage profit per shirt.

$$\begin{aligned} CP &= \$15 \\ SP &= \$20 \\ P &= \$5 \\ \%P &= 33.33\% \end{aligned}$$

- b) If she sells the last 2 for \$10 each, what is the percentage loss for each of these 2 shirts?

$$\begin{aligned} L &= \$5 \\ \%L &= 33.33\% \end{aligned}$$

- c) What is the total percentage loss or profit for all 10 shirts?

$$\begin{aligned} CP &= \$150 \\ SP &= \$180 \\ P &= \$30 \\ \%P &= 20\% \end{aligned}$$

**Question 14**

(5 marks)

Marco owns a store. He makes a profit by selling his material for more than his purchase price.

Complete the following table.

| Cost Price | Sell Price | % increased by (the markup) |
|------------|------------|-----------------------------|
| \$16       | \$19.20    | 20%                         |
| \$55       | \$63.25    | 15%                         |
| \$100      | \$117      | 17%                         |
| \$5        | \$5.50     | 10%                         |
| \$29       | \$35       | 20% / 21%                   |

**Question 15**

(4 marks)

420 reduced by 90% can be calculated using which 4 formulas below (circle all the correct formulas)

a)  $420 \times 9 - 100 =$

b)  $90 \div 420 \times 100 =$

c)  $420 - 420 \times 90 \div 100 =$

d)  $420 \times 10 \div 100 =$

e)  $420 - 90 \div 100 \times 420 =$

f)  $420 - 90 \div 100 \times 420 =$

g)  $(420 - 420 \times 90) \div 100 =$

h)  $420 \times 0.1 =$

**End of Test**